

Junkers and Rohrbach set up factories in the Soviet Union where they produced military aircraft prototypes in defiance of the peace treaty. At the same time, the transfer of technology from Germany boosted the development of Soviet military aviation, which became especially dependent on German aero-engines.

American isolationism

In the United States, the 1920s were a lean period for military aviation. In a reaction against America's involvement in World War I, public sentiment was overwhelmingly "isolationist." Determined to keep out of foreign quarrels, Americans saw their military needs as purely defensive. Since the only credible threat to the United States was an attack from the sea, the navy received the lion's share of a much reduced military budget. Advocates of a powerful independent air force with equal status to the army and navy had a hard furrow to plow, taking on both the chiefs of the established services – eager to keep control of their own air forces – and the general perception of America's defense needs.

The leading advocate of an independent US air force, General Billy Mitchell, made what was under the circumstances a pretty effective job of advancing his cause. The argument for aviation as the offensive arm that would win a major war had little impact, since that was not the kind of conflict the United States intended to get involved in. So Mitchell took it upon himself to demonstrate that aircraft could take over responsibility for the defense of America's coastal waters. The idea won some backing in Congress after Mitchell pointed out that a large fleet of bombers could be built for the price of a single battleship. The navy was forced to allow Mitchell the chance to prove his point.

Mitchell's demonstrations

In July 1921 Mitchell assembled some Martin MB-2 bombers and, in front of naval observers, undertook to sink three German warships that had come into American hands at the end of the war. Armed with 600-lb (270-kg) bombs, the Martins made short work of a destroyer and a light cruiser. The key test was whether they could sink the third vessel, a captured, heavily armored battleship, the *Ostfriesland*. Their first attempt failed. The following morning, a series of attacks with 1,100-lb (500-kg) bombs left the battleship damaged but still afloat. Finally, a strike by seven bombers carrying 2,000-lb (900-kg) bombs sent the battleship to the bottom of the sea. Mitchell was triumphant and some naval observers reportedly watched with tears in their eyes as the pride of

WILLIAM "BILLY" MITCHELL

WILLIAM "BILLY" MITCHELL (1879–1936) earned rapid promotion in the US Signal Corps in the years leading up to World War I. In 1912, aged 32, he became the youngest officer ever posted to the General Staff. Mitchell was an early enthusiast for aviation and in 1917 emerged as the leading combat commander of the US Army's air forces on the Western Front. He struck up a close relationship with RAF chief Sir Hugh Trenchard and became, like Trenchard, an advocate of independent air power. As assistant chief of the Army Service, Mitchell campaigned tirelessly for a well-funded US air force, lobbying Congressmen, writing popular

books and articles, and staging publicity stunts such as the sinking of the *Ostfriesland* and the flight of the Douglas World Cruisers.

But Mitchell's vigorous self-promotion and his public denigration of senior officers who disagreed with him went far beyond what was acceptable from a serving officer. In 1925 he was court-martialed and suspended from duty, after accusing his superiors of "incompetence [and] criminal negligence." Mitchell has remained a controversial figure in America, regarded by many aviation enthusiasts as an inspired prophet of air power.



HIGHLY DECORATED COMMANDER

The decorations and medals of General William "Billy" Mitchell, recognized as the top American combat commander of World War I, included the Distinguished Service Cross and Medal, several foreign decorations, and a posthumous Congressional Medal of Honor for outstanding services to military aviation.



the oceans succumbed to air power. Others must have reflected on whether it was in fact so impressive to sink a tethered and undefended warship at the third attempt.

Mitchell's efforts may have had some effect in advancing the cause of naval aviation, but his

"In the development of air power, one has to look ahead and not backward and figure out what is going to happen..."

WILLIAM "BILLY" MITCHELL

advocacy of an independent air force came to nothing. The Army Air Service was officially upgraded to the Army Air Corps in 1926, but remained in practice an underfunded, subordinate branch of an underfunded army – during the 1930s, United States land forces were smaller than those of Poland or Romania.

The deficiencies of the Air Corps were publicly revealed when the army briefly took over flying US airmail routes in the winter of 1934 (see pages 114–15). The army aircraft – mostly

small, open-cockpit biplanes – were inferior to civil aircraft, and army pilots generally had little or no experience of flying in bad weather conditions or at night. In short, American military aviation had fallen years behind the most advanced civil aviation. Air Corps pilots

averaged more than one accident for every 1,000 hours of flying time. It was normal for them to die at the rate of about one a week. By bringing the spotlight of publicity to bear on these deficiencies, the attempt to fly the mail led to some improvements in equipment and training.

Seaborne aviation

Naval aviation in the United States made better progress, despite financial stringencies. The example of Britain suggests that this progress may have been partly due to the absence of an independent air force. For, at the end of World War I, Britain had led the way in the development of carrier-borne aviation. HMS *Eagle*, which joined the Royal Navy in 1924, set the template for future aircraft carriers, with an "island" set to one side of the flight deck incorporating a bridge